

The Ephemeris

August 2013

Volume 24 Number 7 - The Official Publication of the San Jose Astronomical Association.



Houge Park August Events

8/2

Beginner Astronomy Class 8:15 p.m.
Star party. 9:15 p.m. - midnight.

8/4

Solar observing. 2 - 4 p.m.
Fix-It Day 2 - 4 p.m.

8/16

Star party. 9:30 p.m. - midnight.

8/17

6 - 7:30 p.m. Board of Directors Meeting.
General Meeting
7:30 - 8 p.m. Social time.
8 - 10 p.m. Speaker: Dr. Chris McKay.

8/30

Beginner Astronomy Class 7:45 p.m.
Star party. 8:45 p.m. - midnight

From the Editor - Mina Wagner

Greetings Earthlings! As we go into August some of you might get that sinking feeling that our observing days are numbered. But don't despair. We will continue to observe well into October with CalStar being October 3, 4 and 5.

I have a couple of great success stories for this month.

The SJAA is now the proud owner of a SUPER LARGE screen in our Houge Park venue. It saw first light on Saturday the 20th when the SJAA showed a double feature: Forbidden Planet, and Star Wars, episode IV (the first one released, the original, and the best in many people's minds). It was SO much fun! We had plenty of free popcorn and lemonade. Some people brought their own chair (a great idea because you don't want to sit on our regular chairs for that long. I'm bringing my chaise-lounge next time). Also, next time we need to have a dessert potluck!

Here's a question for all of you...how often would you like to have movie-night? 2x per year? 3x? 4x? 1x?

We had the Golden Gate Star Party at the beginning of July. I heard it was a great success. (Unfortunately I wasn't able to go this year). I have included a couple of articles with photos.

SJAA Contacts

President: Rob Jaworski
Vice President: Lee Hoglan
Treasurer: Michael Packer
Secretary: Teruo Utsumi
Director: Rich Neuschaefer
Director: Greg Clayton
Director: Dave Ittner
Director: Mina Reyes Wagner
Director: Ed Wong

Beginner Class: Mark Wagner
Deep Sky: Mark Wagner
Fix-it Program: Ed Wong
Imaging SIG: Harsh Kaushikkar
Library: Dave Ittner
Loaner Program: Dave Ittner
Lunar/Planetary: Akkana Peck
Ephemeris Newsletter -
Editor: Mina Wagner
Production: Mark Wagner
Publicity: Rob Jaworski
Questions: Lee Hoglan
Quick STArT: Dave Ittner
Solar: Michael Packer
School Events: Jim Van Nuland
Speakers: Teruo Utsumi

E-mails: <http://www.sjaa.net/contact.shtml>



Big Screen at SJAA!

SJAA's new big screen has debuted :

"I had not seen "Forbidden Planet" since its original release, in 1956. It is amazing how influential that movie was; looking at the film I could see elements that subsequently appeared in "Star Trek", "Star Wars", the "Alien" series, and perhaps even in "Avatar"; furthermore, "Forbidden Planet" is in turn loosely based on Shakespeare's "The Tempest" (featur-

ing Robby the Robot as the sprite, Ariel). It is still worth seeing, even after more than 50 years development of special effects and of real spaceflight. (Most of the specials must have been hand-animated, they would have been incredibly expensive at today's prices.) The original "Star Wars" is better known, and is also on most people's one-of-the-best list -- justly so.

Thanks to the presentation team and to the SJAA for the beverages and the popcorn. I hope I was not the only person to put money in the kitty!"

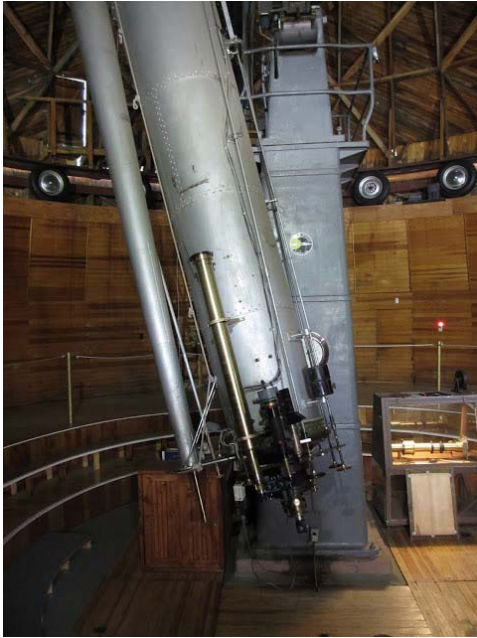
Jay Reynolds Freeman

In this issue:

From The Editor	Pg 1	Moon Targets for Houge Star Parties	Pg 8
Houge Park Events	Pg 1	Deep Sky Targets for Houge Star Parties	Pg 8
Grand Canyon Star Party	Pg 2	SJAA Board of Directors	Pg 9
Golden State Star Party	Pg 4	School Star Party Program Success!	Pg 9
There Are No Vulcans in the Underworld	Pg 7	Heard and Overheard	Pg 9

Grand Canyon Star Party, June 2013

By Marilyn Perry



Clark 24" Refractor Telescope

I expected that the highlight of my June trip to Arizona would be to attend two nights of the 23rd Annual Grand Canyon Star Party at the South Rim. Just in case of clouds at the party, I packed my 10x50 binoculars, tripod, and camp stool, so I could go northern Arizona star gazing every clear night. I also knew the moon would have a bigger impact each night. The Star Party started at New Moon, June 8, 2013, but it lasts 8 days, so the moon would be nearly First Quarter by the time I got to the Star Party.

I was taking a Road Scholar program, an educational tour consisting of astronomy lectures by professors of Northern Arizona University at Flagstaff, field trips to Lowell Observatory and Meteor Crater and culminating with the Grand Canyon Star Party. There were a few star gazing enthusiasts like me (one couple drove from Tennessee so they could bring their 8" telescope), but there were also folks from Manhattan and Brooklyn who said they never got to see stars, and just wanted to see some stars.

The first night in Flagstaff was cloudy. The next day in Flagstaff we toured the Lowell Observatory on Mars Hill, elevation 7,200 feet, very conveniently located just about one mile west from the historic Flagstaff downtown. We saw the Clark Telescope, a 24" refractor which Percival Lowell had built in 1896 so that he could study the "canals" on Mars. The lens was designed by Alvan Clark.

This telescope is now used only for public outreach. It is open for daytime tours and, if weather conditions allow, summer nighttime viewing. Observe the flat panels in the sides and roof of the dome which was built of locally grown ponderosa pine. The dome rotates on tires which replace the original worn out metal wheels. The roof doors are flat panels that open outward, and turn

into sails in high winds. I did not get to view through this telescope due to the wind.

We also saw the Pluto Discovery Telescope which was built after Lowell's death in 1916 with funds provided by his younger brother to continue the search for "Planet X" which Lowell had begun in 1905.

Clyde Tombaugh used this astrograph with 13" lenses to find Pluto. He used 14" x 17" glass photographic plates to take one hour exposures of a portion of the sky, then photographed the same portion six days later, and compared the plates. He discovered Pluto by comparing plates from January 23 and January 29, 1930.

The Lowell Observatory is a non-profit research institution, and it runs several other telescopes which are outside the Flagstaff city limits, including the new \$53 million, 4.3 meter Discovery Channel Telescope located 40 miles southeast of Flagstaff.

That evening was very clear, but I could not view through the Clark, so I did the best I could on my own, even without a car. Flagstaff is designated as an "International Dark Sky City", but there were still lots of streetlights around the motel. It is located at the end of a street which dead ends at a little hill with no lights to the north, but lots of street lights on all other sides. I put up the hood of my sweatshirt, pulled it out as far as I could to block the light from the sides, looked to the north through my binoculars, and, oh my goodness, so many stars!!! Even with all these street lights around me, I could just glance up and see the Coma Berenices Cluster. Walking back to the motel, I could see Scorpius shining brightly, so I decided to try to see globular cluster M4. I thought it would be futile because I was looking straight into the street lights, but I gave it a try. To my delight, with my binoculars I could see M4 very clearly. Flagstaff skies at 7000 feet are so different from Silicon Valley skies.

The next day was our field trip to Meteor Crater, Winslow, AZ, located about 40 miles southeast of Flagstaff. It was formed when an estimated 45 meter meteorite hit about 50,000 years ago. The crater is nearly one mile across, 2.4 miles in circumference, and about 550 deep.

At first the crater was thought to be of volcanic origin. In 1902 mining engineer Daniel Moreau Barringer thought it was caused by the impact of a large iron meteorite. He spent the rest of his

(continued next page)



Pluto Discovery Telescope



Panorama of Meteor Crater

(Grand Canyon Star Party continued)

life until his death in 1929 mining in the crater in an unsuccessful attempt to find the iron meteorite. He could not find it because it did not exist—it had been broken up into tiny pieces by the impact. Barringer did live long enough to see the scientific community start to accept his theory that the crater was formed by a meteorite.

Impact craters have been called “nature’s drills” because they sort of invert the strata. Apollo astronauts trained at Meteor Crater in preparation for walking on the moon because Meteor Crater is so much like craters on the moon. By recognizing impact craters on the moon, the astronauts could collect samples of rocks originally from below the surface without having to drill for them.

Then it was on to the South Rim of the Grand Canyon. The Grand Canyon Star Party began as the first anniversary of Dean and Vicki Ketelsen’s honeymoon there, and telescopes set up by the Canyon proved so popular with tourists, that the event became an annual tradition as a public oriented event. Skies are typically clear in June at the Grand Canyon whereas monsoon season typically begins later in the summer.



Mercury and Venus over Grand Canyon

Arizona does not observe Daylight Savings Time, so skies get dark earlier in the evening at the Grand Canyon than in Silicon Valley. The Tucson Amateur Astronomy Association hosts the Star Party on the South Rim and the Saguaro Astronomy Club based in Phoenix hosts the Star Party on the North Rim.

Before the party began I took this photo of Venus and Mercury above the Grand Canyon from Mather Point. Mercury is barely visible at a 45 degree angle to the left and above Venus. I photographed this scene over and over as the sky darkened. People asked me what I was doing, and I got to point out Mercury to many people who had never seen it before. Using Venus as a guide, everyone was eventually able to spot it.

Sitting with my binoculars at Mather Point watching the canyon darken and the stars come out was absolutely the highlight of my trip. The moon was out, and I could see my shadow clearly, so I felt no fear that I would accidentally fall into the canyon. Also there are big strong metal railings at Mather Point. By all the green laser lights pointed to the sky, I could see that the Star Party had begun, but I did not want to leave the glorious vista of horizon to horizon bright stars above and the moonlit canyon below with only one small light at Phantom Ranch. It was hard to do, but eventually I pulled myself away from the canyon and walked in the moonlight and the light of my flashlight to the Star Party. By the time I got there, the crowds had thinned, and I got to see fabulous views through the big telescopes.

The next morning I was up early for hiking and exploring. Members of the Tucson AAA set up for solar viewing right on the South Rim between the



Solar Viewing at Grand Canyon South Rim

shuttle bus stop and the Bright Angel Trail head. I got to see solar prominences, but no sun spots. The weather was beautiful, and the daytime sightseeing spectacular, especially hiking along the Rim Trail and enjoying many viewpoints along Hermits Road. Not having a car was no problem due to the excellent free shuttle buses.

Then it was time for the second night of star gazing at the Grand Canyon Star Party. The amateur astronomers set up their telescopes in the employee parking lot at the Grand Canyon Visitor Center at Mather Point. The area is several hundred feet from the rim, so there is no danger of anyone falling into the canyon.



Setting up telescopes for Grand Canyon Star Party

(continued on next page)

(Grand Canyon Star Party continued)

I was very impressed with the large reflecting telescope whose mirror you can see to the right of the tall ladder. The view of the M13 globular cluster from that telescope was fantastic! To me it looked just like the photo of M13 in the Sky Safari star gazing app.

In this photo the ranger talk was just over, the crowds were arriving, and the Star Party had just begun. I took this photo with ISO 1600, exposure time 1 sec, so the sky appears light, but it was actually quite dark, and the astronomers were sharing very good views of Mercury and Saturn. I estimate there were forty telescopes, mostly set up by amateur astronomers from Tucson, although I spoke to one who made an almost annual trip from Texas. At first the crowds were heavy, and the lines at the telescopes were quite long, but they thinned out after 10 PM. The evening also included a bright Iridium flare, a very nice pass by the International Space Station, and hourly green laser Constellation Shows, all announced ahead of time by the astronomers.



Star Party has just begun

I had a wonderful time, and I especially loved sitting at Mather Point watching Mercury and Venus sink while the canyon got dark and the stars appeared. One thing that I missed was watching the Milky Way rise over the canyon. I would like to go back in the dark of the moon so I could see that.

For more photos and information about Lowell Observatory, Meteor Crater, and the Star Party, see my blog posts on the SJAA blog: <http://sanjoseastronomy.blogspot.com/>

Golden State Star Party Report

By Michael Packer

For the 6th year the GSSP was held 100 miles northeast of Redding at 4300 feet on flat ranch land near the town of Adin. This year was probably the best weather we 400 observers and imagers had to date.



Left: Entrance to Star Party And Mt. Shasta. Right: View From My Campsite With Venus.

The Crescent Nebula, NGC 6888, which usually looks like its name in scopes 12" or larger looked more like a ghostly blue "Vincent van Gogh's ear" with OIII filter. Lots of wispy detail inside and of course with these skies the Veil and Swan Nebulae were soaring. Curiously, the 4 nights seemed to offer an opportunity to reinvestigate NGC's in doublets, triplets and even quintuplets. My favorite globular cluster pair NGC 6528 & 6522 just 0.25 degree apart were sparkling but M51a/b were spiral banded and bound even in scopes 9-10". The Draco Triplet offered an exquisite lesson in classic galaxy type in moderate scopes. But in larger scopes the number of interacting galaxies particularly in Virgo were hot targets. I stopped by a 28" scope to look at NGC 4567 and NGC 4568. Known as the Siamese Twins or the Butterfly Galaxies, the view was revolutionary. The galaxies looked exactly like a recent Hi-Def N-body computer simulation I had seen of two galaxies just at the moment of collision. I got the sense of a massive number of stars being stripped away by tidal forces yet this view was perfectly narcotized by time and space and perfectly real. Stephan's Quintet was superb x5!

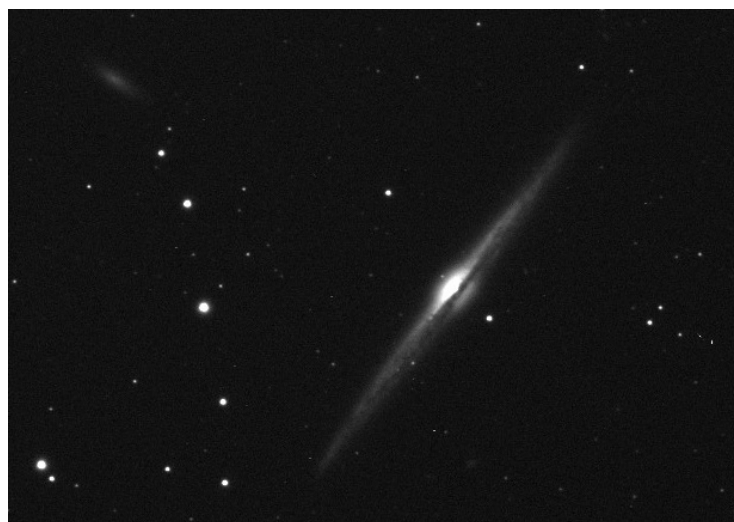
(Continued on next page)

(Golden State Star Party Report continued)

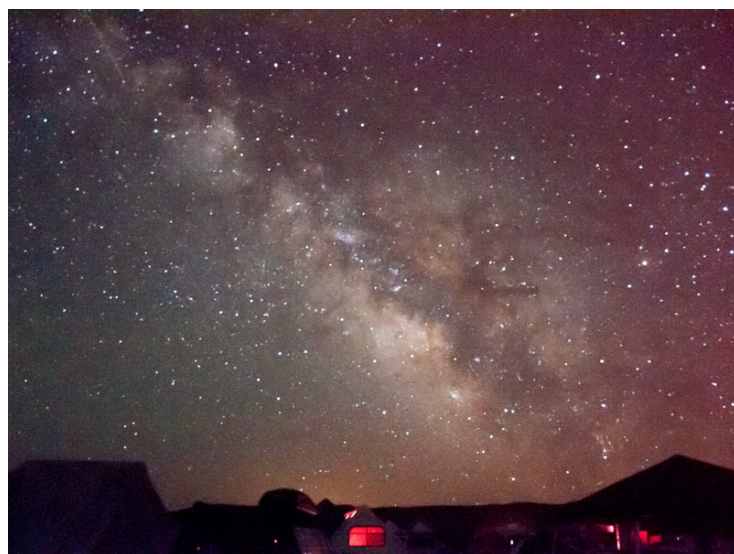
SJAA had a good turn out of members. Below are San Jose folks Rudy Prater, Andrew Thomas, Michael Packer, Bill and Susan O'Neil.



And below is SJAA member Mark Scrivener (Left) with Hitesh Dholakia and his son Shishir. Both Hitesh's sons Shashank and Shishir came along. I believe Gregory Edwards was also present and I even heard Harsh Kaushikkar stopped by.



The above (Right) image of NGC 4565 was taken by Scrivener. Image details: 2 stacked 300sec frames post-processed using Nebulosity3. Equipment: 90mm, F/6.2 with Orion StarShoot Deep Space Monochrome III. He also took the below (Left) image of the Helix Nebula (NGC 7293) similarly processed with 6 stacked 300sec frames.



The above (Right) image of the Milky Way was taken by Susan O'Neil. Details: 15sec ISO5000 tripod mounted Nikon D300 using an f/3.5 18mm lens.

(Continued on next page)



Lots of cool scopes were at GSSP. Seen above, an Albert Highe compound dob, completely hand made refractor and spectrohelioscope.



Local town folks and Adin Supply served one of the three dinners star party goers could reserve.

On a final note the three dinners prepared by the local town folks were also pure clear sky heaven. My favorites were the Tri-tip, wild rice and chicken tamales. The tamales were surprising as I am accustomed to real Tex-Mex made fresh on my grandfather's ranch in the Panhandle.

Shallow Sky - There Are No Vulcans in the Underworld

By Akkana Peck

August skies are ruled by the outer solar system.

Saturn is an evening object, though it won't be very high, transiting before nightfall and setting a little before midnight. So catch it early, before it gets too low.

Venus is also visible in the evening sky; it sets around 10 pm at the beginning of August, edging closer to 9 by month's end.

Mars and Jupiter are morning objects this month, in Cancer and Gemini respectively. Mercury, too, is a morning object, but it hugs the sun all month and won't be an easy target.

But the real show this month is in the outer solar system. Uranus, Neptune and Pluto are all well placed for observing. Uranus rises late and doesn't transit until after 4 am.

Pluto, transiting about 10:30 pm, is still in the extremely star-rich field of northern Capricornus, 4 degrees north of M22 and a little under 2 degrees southwest of M25. You'll want a good, very detailed finder chart, like the one in the RASC Observer's Handbook, to identify it, or a reliable star charting program.

Since not all planetarium programs calculate Pluto's position correctly, if you want to be sure you've spotted our distant neighbor, make a sketch of the star field, indicating the one you think is Pluto; then wait a few days and find it again. Pluto doesn't move fast, but you should be able to see motion if you observe regularly a few days apart over a week or two.

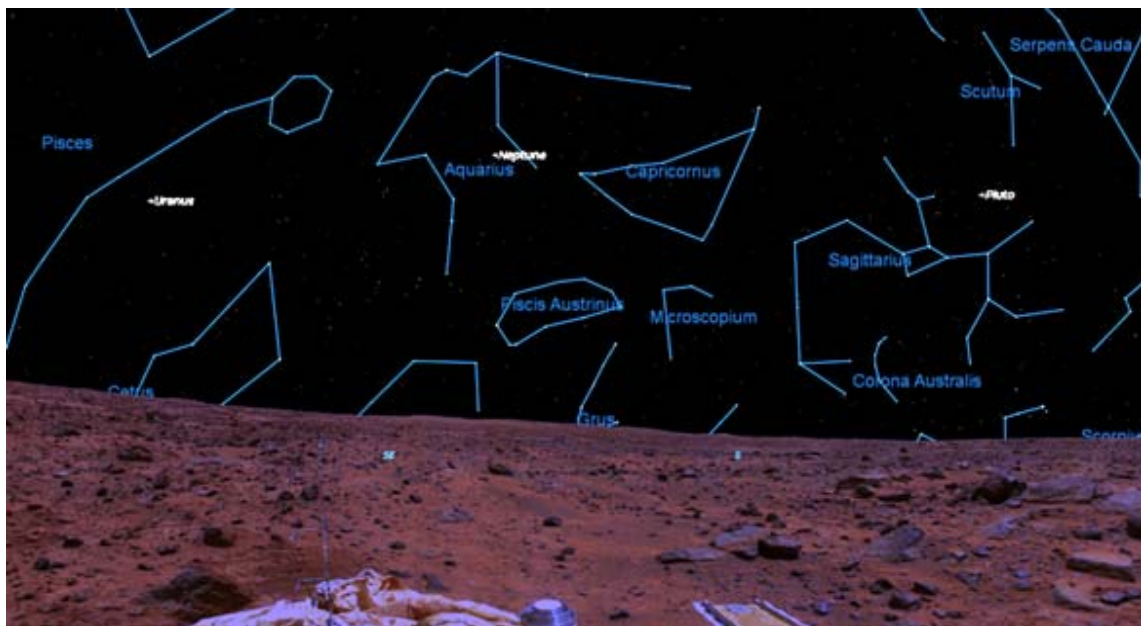
Once you're pretty sure you have Pluto, try to spot its moons. Ha! Just kidding. Pluto's moons have been in the news recently, but

Hydra, are quite a bit smaller, estimated at around 30-40 miles in diameter. But Pluto has two even smaller moons, and at about the same time New Horizons was shooting Charon, the SETI institute here in Mountain View announced the results of a naming contest they'd run for Pluto's two tiniest moons. (Why did the SETI institute get to name the moons? I haven't found any reportage that explains that -- maybe they just ran the contest hoping that the International Astronomical Union, which is responsible for naming planets and moons, would listen to the results.) Almost 500,000 people voted in the contest, and the winner by a landslide was Vulcan. (Leonard Nimoy, Star Trek's original Vulcan, reportedly tweeted that the name Vulcan was "the logical choice".)

But the IAU vetoed the name Vulcan, insisting on names in keeping with the

mythological theme of Pluto, god of the underworld. It accepted the second and third place names, Cerberus and Styx. In Cerberus' case, the Plutonian moon will use the Greek variant of the name, Kerberos, instead of the Latin Cerberus, since there's already an Apollo asteroid named 1865 Cerberus. Cerberus/Kerberos, of course, was the three-headed dog guarding the gates of the underworld.

Kerberos has an estimated diameter of 8-21 miles, while Styx is thought to have an irregular shape of about 6-15 miles across. I'm sure we'll know a lot more about them once New Horizons gets there in 2015.



It's in Pisces, in a guide-star sparse area off the tail of Pegasus.

Neptune, at opposition on August 27th, is in Aquarius halfway along the line between Ancha (Lambda Aqr) and 4th magnitude double star Tau Aqr. That might make it relatively easy to find, so take a shot at it if you haven't seen the blue planet in a while. At least, I always see Neptune as blue compared to Uranus' green. But people see colors very differently, so I'm sure some people see them as both blue, both green, Neptune green and Uranus blue, or no color at all. I'd be interested in hearing what colors other people see.

not because of being spotted in amateur telescopes.

The New Horizons spacecraft, speeding toward its eventual rendezvous with the tiny planet, just resolved Pluto's largest moon Charon for the first time on July 1. Charon is about 750 miles in diameter, roughly the size of Texas, and orbits Pluto at a distance of about 12,000 miles -- a lot closer than our moon's 239,000 miles. From where New Horizons is now, 550 million miles from Pluto, that's about half an arc minute of separation, so not too bad for the spacecraft's on-board camera.

Pluto's second and third moons, Nix and

Did you look for Pluto and find it too easily? Want another challenge? In mid August, ambitious hunters can start looking for Comet ISON, the comet everybody's hoping will give us such a great show this winter. This month it emerges into the dawn sky in Cancer. It's only 11th magnitude, so it's quite a bit brighter than Pluto's 14th magnitude. On the other hand, it's a comet, so it'll be at least somewhat fuzzy and that 11th magnitude may be deceptive. In any case, it should be within reach of a dedicated observer with a large telescope. So if you're straining at the bit to get an early start on this season's big comet, now's your chance!

Moon Targets for August 16 Houge Park Star Party

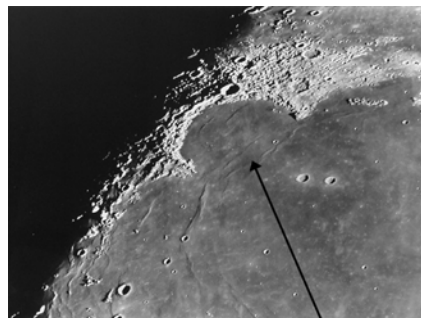
By Mark Wagner

Here are four targets that should be reasonably well placed for observing. It would be helpful to find someone with a Rukl Lunar Atlas, or to have a lunar map for reference. A facsimile of the Rukl Atlas is available on-line at <http://the-moon.wikispaces.com/Rukl+Index+Map>. Lunar images courtesy of the Lunar and Planetary Institute.



Enke - Rukl pg 30. 4.6N, 36.6 E.

Dave North notes: "Encke is one of the most hexagonal craters on the moon."



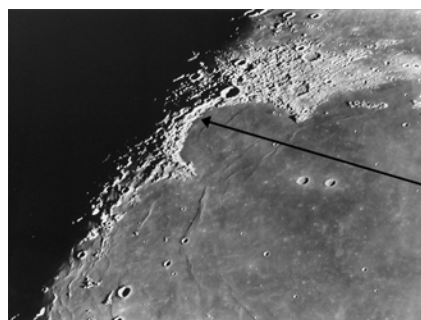
Sinus Iridium - Rukl pg 10. 45.0N, 32.0W.

Jay Reynolds Freeman: "Note the long, prominent wrinkle ridges that cross the mouth of Sinus Iridum and trend southward, past the crater Caroline Herschel, through Lambert and all the way to the Carpathians"



Kepler - Rukl pg 30 8.1N, 38.0 W

Akkana Pecks notes say: "Kepler has a terrace in its wall, very prominent when it's near the terminator."



Montes Jura - Rukl pg 10 47.0N 37.0W

"The mountain range borders Sinus Iridum as the wall of a flooded crater."

(Atlas Of The Moon—Rukl)

Deep Sky Targets for August 2 and August 30 Houge Park Star Parties

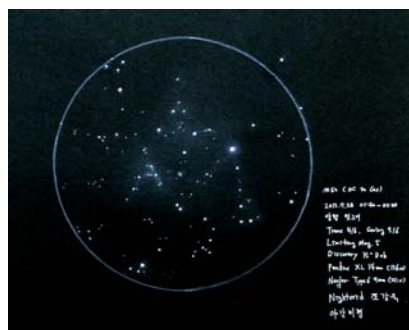
By Mark Wagner

The moon sets early at this Houge Star Party, so here are some deep sky targets. Even in brighter suburban skies, one can glimpse deep sky targets and have satisfying views. Drawings from Astronomy Sketch of the Day: <http://www.asod.info/>



M11 - Open Cluster
Scutum
RA 18h 51m 52s
Dec -06°14'46"
Mag 5.80

The "Wild Duck" is rich cluster is bright and easy to find. Many stars of nearly the same magnitude punctuated by a few very bright members. One of the great open clusters.



M52 Open Cluster
Cassiopeia
RA 23h 24m 53s
Dec +61°39'32"
Mag 6.90

Cluster is large and coarse, rich in stars, much compressed middle, round, stars of magnitude 9 to 13;



NGC 7662 Planetary Nebula
Andromeda
RA 23h 26m 36s
Dec +42°37'35"
Mag 9.00

This is called the Blue Snowball, and its color should tell you why. Its small size makes its mag 9 bright from Houge Park. Look closely and see if you can note its "annularity" - a darkening of the central part



Epsilon Lyrae - Double star
Lyra
RA 18h 44m 49.219s
Dec +39°41'26.436"
Mag 4.67

You'll see this from Houge Park without optical aid. But with sufficient magnification, this "double star" will reveal itself a not on double, but a pair of double stars. Really, try it!

SJAA Board of Directors

Donation!

From SJAA President Rob Jaworski:

I'd like to mention a bit of feedback about the SJAA that came in during the movie night last Saturday.

A relatively new member (about 3 months or so) came to me and expressed his appreciation for all the things that the SJAA does for the community. He really seemed to most enjoy and appreciate the in-town star parties that we host on a frequent basis, and he also was fond of the other events that help keep up the interest of astronomy by the general population. He decided to back up his appreciation with an on-the-spot cash donation of \$100.

Thanks again, everyone, for all that you do. It does NOT go unnoticed!

SJAA School Star Party Success!

Congratulations to Jim Van Nuland on another outstanding year bringing astronomy to schools throughout the south bay area.

Final list of school events for 2012-13 school year

Completed events as of June 18, 2013

- A) Total Scheduled
- B) Good sky
- C) Part. success
- D) Cloudy fail
- E) Cancel at noon

Start 2012-13 school year					
	A	B	C	D	E
Jul	2	2	-	-	-
Aug	1	1	-	-	-
Sep	1	-	1	-	-
Oct	7	3	-	-	4
Nov	11	4	-	-	7
Dec	13	4	-	-	9
Jan	8	7	-	1	-
Feb	8	5	-	-	3
Mar	6	2	1	-	3
Apr	5	4	1	-	-
May	3	3	-	-	-
End 2012-13 school year					
Tot	65	35	3	1	26

HEARD AND OVERHEARD... By Mina Wagner



There is already chatter on e-mail lists about the CalStar dark sky star party. Are you planning on going? It occurs the nights of October 3,4 and 5, at nearby Lake San Antonio. This is a very dark site within 2-1/2 hours of San Jose!

I hear the SJAA is looking into acquiring a dark sky observing site. Discussions have been taking place at the club's monthly board meetings.

Coming soon to the SJAA, an updated web-page, using a Content Management System. This will put management of the site into more hands - those managing programs will be able to directly input information onto the web-site with his "CMS". It will allow additional flexibility for the club and its membership. Who knows, perhaps you'll want to be involved.

I've heard that the SJAA's great Quick STARt program has had its budget increased, to purchase additional equipment. The program gets newcomers to the hobby up and running, with loaned equipment and instruction. Demand has been good and steady.

There is also a new Binocular Astronomy program, headed up by board member Ed Wong. The next program will take place in August 2 at Rancho Canada del Oro. Using binoculars is a great way to familiarize yourself with the sky.

Did you know the SJAA had donated to it World War II Japanese battleship binocular? The club is inquiring into its history - board member Teruo Utsumi has contacts in Japan to see if this pair has some historical value. The donor is a life-member of the SJAA (and two-time club president).

Another board member, Michael Packer, has recommended the board have an off-site brainstorming session - to consider new ideas and directions for the club. It is an alternative to the regular board of director meetings, which tend to follow a familiar format. Who knows where it will lead, but that's part of the FUN of doing it! While having it in Princeville, Kauai, it looks like we'll settle this time for a session at Adobe Systems in downtown San Jose, with a brainstorming facilitator leading us. We'll keep Kauai in mind for the future!

Ed Wong wrote about the SJAA's "Fix-It" Program (each first Sunday of the month): "The Fixit session this past month we had a gentleman bring in a replica of Newton's telescope that he had built. He was having some questions about the mirror and the aberrations in it. Dave and Phil looked at it and answered his questions. We had 2 gentleman Cassegrain scopes that needed help with them. One was a Schmidt-Cassegrain that had a loose secondary mirror that Phil helped him fix. The second on was a Maksutov-Cassegrain that needed cleaning and a visual back adapter. Phil also helped him with that. I worked on collimating the QuickStarT dobs that were returned."

Did you know the Fremont Peak Observatory Association is holding its 28th Annual Fremont Peak Star-B-Que on August 10? From their web-page (www.fpoa.net):

"Potluck Picnic, Star Party, and Annual Meeting on Saturday, August 10, 2013.

Reserve the date. We usually start around 5pm. There will be activity from the annual Perseid meteor shower that night, even though the peak activity is usually around the 11/12."

The Star-B-Q is a great place to find astronomy friends you rarely get to see.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

San Jose Astronomical Association Membership Form

P.O.Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

- Membership Type:**
- ☐ Regular — \$20
 - ☐ Regular with Sky & Telescope — \$53
 - ☐ Junior (under 18) — \$10
 - ☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at**
<http://ephemeris.sjaa.net>.

Questions? Send e-mail
sjaamemberships@gmail.com

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to “SJAA”, or join/renew at <http://www.sjaa.net/membership.shtml>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____